

Prakrititz Borah

Bangalore, India

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Education

International Institute of Information Technology Bangalore

Jul 2023 – Jul 2028

Integrated MTech (B.Tech + M.Tech) in Computer Science and Engineering

CGPA: 3.35/4.00

- Minor in Artificial Intelligence and Machine Learning

Work Experience

Research Work | [\[Paper, BIOSTEC 2026\]](#)

Oct 2025 – Dec 2025

Samsung Labs, IIIT-Bangalore. Advisor: Dr. Sakshi Arora (PhD, Stanford)

- Ran a wearable-ECG data collection study across many subjects under a standardized 4-phase stress-induction protocol (Baseline, Stroop, Mental Arithmetic, Recovery), producing ~20 hours of labeled physiological recordings for the study.
- Built the signal-processing pipeline from raw sensor output: reconstructed continuous ECG from timestamped device packets, filtered noise, and extracted HRV features across time, frequency, and nonlinear domains – turning noisy consumer-grade hardware into research-usable signal.
- Implemented a **Genetic Algorithm** to search the HRV feature space for the most informative subset, cutting the feature count nearly in half while *improving* downstream classification accuracy over the full feature set.
- Designed and trained **CTBFNet**, a hybrid CNN–Transformer–BiLSTM model with HRV-guided attention, then benchmarked it against lightweight ML baselines to map out the accuracy-vs-compute trade-off between the two approaches.

Project Intern [\[Report\]](#)

Apr 2024 – June 2024

IIT-Bombay, E-yantra program

- Formulated a **location-aware circadian lighting algorithm** utilizing Gaussian, sinusoidal, and parabolic models to accurately approximate natural color temperature variations.
- Integrated **geolocation and sunset-sunrise APIs** to dynamically synchronize lighting transitions with the user's local day-night cycle.
- Developed a hardware prototype using an **ESP32 microcontroller** and **NeoPixel RGB LEDs**, and built a **Flask web interface** for manual RGB adjustments and automated circadian rhythm simulation.
- Implemented complex mathematical transformations to convert correlated color temperature (CCT) into **CIE XYZ** and **sRGB** color spaces for precise digital color rendering.
- Integrated the system with the **Tuya IoT platform** to ensure high scalability and enable seamless deployment across existing smart home ecosystems.

Publications

Stress Detection from Consumer-Grade ECG Device Using GA Optimization

Mar 2026

19th International Joint Conference on Biomedical Engineering Systems and Technologies | [link](#)

Marbella, Spain

Projects

RelayBrain | [Node.js](#), [MCP](#), [Multi-Agent Systems](#) | [link](#)

May 2026 – Jun 2026

- Designed and open-sourced a **cross-agent memory architecture** (npm package + CLI) unifying shared context across **5 heterogeneous coding agents** (Claude Code, Cursor, Copilot, Codex, Antigravity) via a single **versioned knowledge store**.
- Built a **session-boundary extraction pipeline** using per-agent stop hooks to compile raw transcripts into structured **intermediate-representation** snapshots (task state, decisions, failure log), enabling lossless context handoff on agent switch.
- Exposed the shared store as an **MCP server** over stdio (5 structured read/write/sync tools) with a **zero-database, file-only design**; validated across Claude Code, Cursor, and Copilot CLI.
- Implemented an **automatic config patcher** rewriting each agent's native settings to point at the shared brain, plus a **Mission Control dashboard** (Next.js) for real-time cross-agent activity. **6 stars, 4 forks**.

RAG Prompt Compression For Code | [PyTorch](#), [HuggingFace](#), [AST Analysis](#) | [link](#)

Feb 2026 – Apr 2026

- Reimplemented **CodePromptZip** (arXiv:2502.14925) for the Bugs2Fix code-repair task: built a **type-aware AST tokenizer** (JavaParser) with 5 priority token classes and a **priority-driven greedy compression** algorithm.
- Constructed a **471K-pair corpus** (52K Java bug-fix examples × 9 compression ratios) with leakage-safe splitting; fine-tuned a **copy-augmented CodeT5-Base (220M)** pointer-generator over **70K steps (~100 GPU-hrs)** on a single 8GB GPU.
- Built a full **RAG evaluation harness** (BM25 → learned compression → CodeLlama-13B → CodeBLEU/syntax/dataflow scoring) across 9 compression ratios and 0–3 retrieval shots.
- Identified a **non-monotonic compression-quality curve** and a **shot-count collapse effect** (CodeBLEU 91.7 → 4.2 beyond 1-shot) absent from the original paper, isolating prompt-format saturation as the cause. Best: **41% token reduction at 80.36 CodeBLEU**; deployed to Hugging Face.

Multi-Objective NAS on Fashion-MNIST | *PyTorch, Optuna, Bayesian Optimization* | [link](#) Jan 2026 – Feb 2026

- Built a **multi-objective neural architecture search** framework using **Optuna** to jointly optimize accuracy, latency, and parameter count over an **8-dimensional dynamic search space**, using Optuna's **define-by-run API** for conditional hyperparameters.
- Ran a controlled **60-trial study** comparing an evolutionary sampler (**NSGA-II**) against a Bayesian sampler (**MOTPE**) under a fixed seed, measuring latency via explicit CUDA synchronization.
- Quantified Pareto-front quality via **Hypervolume, Spacing, and Spread**: MOTPE found a **26% larger hypervolume** and denser front (24 vs. 18 solutions) than NSGA-II, which achieved wider extreme-point Spread – characterizing the sample-efficiency vs. boundary-exploration trade-off.
- Uncovered a **non-obvious inverse trade-off** between model size and inference speed (a 17K-param model ran *slower* than a 36K-param model), attributed to GPU parallelism favoring wide-shallow over narrow-deep architectures.

Vehicle Detection in Adverse Weather | *YOLOv8, PyTorch, Computer Vision* | [link](#) May 2026 – June 2024

- Co-developed a **YOLOv8-based vehicle detector** for the ICDEC 2024 VDVWC Challenge, targeting detection robustness across weather/lighting on the **AVD-Dataset** (3,200 images: sunny, rainy, night).
- Designed a **3-stage curriculum training strategy** (sunny-only pretraining → full-dataset fine-tuning → short refinement) to improve cross-weather generalization over training on the mixed distribution directly.
- Engineered a **weather-conditioned augmentation pipeline** (mosaic, mixup, copy-paste, HSV/perspective jitter) combined with transfer learning from pretrained YOLOv8 weights.
- Ran systematic **hyperparameter tuning** across loss-gain weights and augmentation magnitudes; achieved **mAP 0.573 / F1 0.52** on validation as an official challenge submission. **4 stars, 3 forks**.

Technical and Scholastic Achievements

- **Amazon Hackathon (2024): 1st Among 100k+ teams**; invited to **Amazon Seattle** for a leadership gathering.
- **Best Paper Candidate – BIOSTEC 2026 (Marbella, Spain)**: Nominated for best paper among international submissions.
- **Max Expert on Codeforces** with peak rating of **1641**. 1000+ problems solved across different platforms.
- **Rank 61 in CodeQuest 2.0** among **3,000+ participants** in a competitive programming contest organized by IIIT Kottayam.
- **JEE Mains (2023)**: Secured **All India Rank (AIR) 9,312** out of 1,100,000+ candidates, placing in the **top 0.7%** nationwide.
- **IOQM (2021)**: Successfully qualified the **Indian Olympiad Qualifier in Mathematics**.

Teaching and Mentorship

Teaching Assistant – ESS 112 Python Programming Aug 2026 – Present (Fall Semester)

International Institute of Information Technology Bangalore

- Mentored 300 students through tutorials, doubt-solving sessions, and assignment evaluations focusing on Python programming logic and best practices.

Teaching Assistant – CSE 102 Data Structures and Algorithms Jan 2026 – May 2026 (Spring Semester)

International Institute of Information Technology Bangalore

- Organized problem-solving sessions and curated algorithmic questions; conducted tutorials for 250 students to strengthen core concepts in data structures and algorithm design.

Teaching Assistant – EGC 102 Digital Design Aug 2025 – Dec 2025 (Fall Semester)

International Institute of Information Technology Bangalore

- Guided students in digital circuit design principles, conducting hands-on tutorials and lab sessions focused on hardware description using **Verilog** and circuit simulation in **LTspice**.

Technical Skills and Tools

Languages: Python | C++ | C | SQL | CUDA

Deep Learning & AI: PyTorch | TensorFlow | Hugging Face Transformers | OpenCV | YOLO

GenAI & LLM Ecosystem: LangChain | LlamaIndex | LangGraph

Classical ML & Data Science: Scikit-Learn | XGBoost | LightGBM | Pandas | NumPy | SciPy

DevOps: Docker | Kubernetes | Git

Leadership and Extracurricular

ICP Crewsphere IIITB Chapter: Established the campus blockchain club, organized hackathons and technical workshops.

Zense (Tech Club), IIIT Bangalore: Conducted workshops, developed the college's official fest website, and organized tech events.

Film Division, IIIT Bangalore: Organized competitions and produced two award-winning films.

Relevant Coursework

Machine Learning: Machine Learning | Multi-Objective Machine Learning | Natural Language Processing with Deep Learning | Network-Based Computing for HPC and AI | Machine Reasoning

Mathematics: Linear Algebra | Calculus | Probability and Statistics | Number Theory | Discrete Mathematics | Automata and Computability | Algorithmic Thinking